**V3 Answers**

1 (a) This is preferable for increased speed/high frequency performance [2] This is because NMOS transistors generally have higher FTEs than PMOS transistors [1]. This is due to mobility of electrons being approximately 3x that of holes [2].

(b) Same as V1.

(d) Different numerical answer (method same).

2 (b, e) Different numerical answer (method same).

3 (b) Different numerical answer (method same).

3 (e) (i) Increase bias current (2) or W/L of M1/M2 (1), or capacitance (1) at node Vout (1).

(ii) Same as V1.